



**Generation Interconnection
System Impact Study Report
for**

Queue Project AF1-292

FIELDS 34.5 KV

8.94 MW Capacity / 14.9 MW Energy

August 2020

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Dominion.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC), has proposed a Solar generating facility located in Greensville County, Virginia. The installed facilities will have a total capability of 14.9 MW with 8.94 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is November 30, 2022. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-292
Project Name	FIELDS 34.5KV
State	Virginia
County	Greensville
Transmission Owner	Dominion
MFO	14.9
MWE	14.9
MWC	8.94
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF1-292 will interconnect with the Dominion distribution system on the Fields 34 kV #446 circuit.

The IC is responsible for securing right-of-way, permits and constructing the proposed attachment line from the solar facility site to the proposed new substation. Attachment 1 shows a one-line diagram of the proposed interconnection facilities. The IC may not install any facilities on Dominion's right-of-way without first obtaining the necessary approval from Dominion Energy.

5 Cost Summary

The costs associated with physically interconnecting the AF1-292 project to the Dominion distribution system will be documented in the two-party Interconnection Agreement between the IC and ITO.

The AF1-292 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$ To be provided in the two-party IA
Allocation towards System Network Upgrade Costs*	\$ 0
Total Costs	\$ 0

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined

that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

Dominion assessed the impact of the proposed Queue Project AF1-292. The project was evaluated as an 8.9 MW Capacity (14.9 MW Energy) injection at the existing Fields 34.5 kV substation in the Dominion Distribution System, for compliance with NERC Reliability Criteria on Dominion Transmission System. The system was assessed using the summer 2023 AF1 case provided to Dominion by PJM. When performing a generation analysis, Dominion's main analysis will be load flow study results under single contingency (both normal and stressed system conditions). Dominion Criteria considers a transmission facility overloaded if it exceeds 94% of its emergency rating under normal and stressed system conditions. A full listing of Dominion's Planning Criteria and interconnection requirements can be found in the Company's Facility Connection Requirements which are publicly available at: <http://www.dominionenergy.com>.

The results of these studies evaluate the system under a limited set of operating conditions and do not guarantee the full delivery of the capacity and associated energy of this proposed generation facility under all operating conditions. NERC Planning and Operating Reliability Criteria allow for the re-dispatch of generating units to resolve projected and actual deficiencies in real time and planning studies. Specifically, in Planning Studies, NERC Planning Event 3 and 6 Contingency Conditions (Loss of generator, transmission circuit, transformer, shunt device, or Single Pole of a DC line followed by the loss of a generator, transmission circuit, transformer, shunt device or single pole of a DC line) allow for re-dispatch of generating units to resolve potential reliability deficiencies. For Dominion Planning Criteria the re-dispatch of generating units for these

contingency conditions is allowed as long as the projected loading does not exceed 100% of a facility Load Dump Rating.

Note that the ITO findings were made from a conceptual review of this project and cost estimate data contained in this document should be considered high level estimates since it was produced without a detailed engineering review. The applicant will be responsible for the actual cost of construction. ITO herein reserves the right to return to any issues in this document and, upon appropriate justification, request additional monies to complete any reinforcements to the transmission systems.

7 Schedule

The schedule for the required Network Impact Reinforcements will be more clearly identified in future study phases. The estimate elapsed time to complete each of the required reinforcements is identified in the “System Reinforcements” section of the report.

8 Transmission Owner Analysis

8.1 Power Flow Analysis

PJM performed a power flow analysis of the transmission system using a 2023 summer peak load flow model and the results were verified by Dominion. Additionally, Dominion performed an analysis of its transmission system and no further deficiencies were identified.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in Dominion’s “Dominion Energy Electric Transmission Generator Interconnection Requirements” documented in Dominion’s Facility Interconnection Requirements “Exhibit C” located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated protection device (circuit breaker, circuit switcher, fuse) to protect the IC’s GSU transformer(s).

2. The purchase and installation of the minimum required Dominion generation interconnection relaying and control facilities as described in the System Protection noted above. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the Dominion Transmission System Control Center.
4. Compliance with the Dominion and PJM generator power factor and voltage control requirements.

The GSU(s) associated with the IC queue request shall meet the grounding requirements as noted in Dominion’s “Dominion’s Facility Interconnection Requirements” document located at:

<https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

The IC will also be required to meet all PJM, SERC, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and SERC audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the Dominion system.

9.3 Power Factor Requirements

The IC shall design its non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the Dominion transmission system.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.1.1 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Back Panel temperature (Fahrenheit)
- Irradiance (Watts/meter²)
- Ambient air temperature (Fahrenheit) – (Accepted, not required)
- Wind speed (meters/second) – (Accepted, not required)
- Wind direction (decimal degrees from true north) – (Accepted, not required)

10.2 Interconnected Transmission Owner Requirements

See Section 3.4.6 “Metering and telecommunications” of Dominion’s “Dominion’s Facility Interconnection Requirements” document located at: <https://www.dominionenergy.com/company/moving-energy/electric-transmission-access>.

11 Summer Peak Analysis

The Queue Project AF1-292 was evaluated as a 14.9 MW (Capacity 8.9 MW) injection at the Fields 115 kV substation in the Dominion area. Project AF1-292 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-292 was studied with a commercial probability of 100%. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
90240042	314435	6SAPONY	230.0	DVP	314282	6CARSON	230.0	DVP	1	DVP_P1-2: LN 2012-B	single	678.68	99.94	100.41	AC	4.03

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42958356	314554	3BTLEBRO	115.0	DVP	304223	3ROCKYMT115T	115.0	CPL	1	DVP_P7-1: LN 2058-2181	tower	93.0	453.18	454.13	AC	1.09
90239691	924510	AB2-100 TAP	230.0	DVP	314583	6LAKEVEW	230.0	DVP	1	DVP_P4-2: 23872	breaker	459.0	106.1	107.67	AC	7.35

11.4 Steady-State Voltage Requirements

To be determined

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed

with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPAC T
90240324	314563	6CLUBHSE	230.0	DVP	924510	AB2-100 TAP	230.0	DVP	1	3144356SAPONY 230 940480 AE2-033 TAP 230 1	operation	375.06	108.21	110.13	AC	7.36
90240153	924510	AB2-100 TAP	230.0	DVP	314583	6LAKEVE W	230.0	DVP	1	3144356SAPONY 230 940480 AE2-033 TAP 230 1	operation	375.06	129.82	131.74	AC	7.35
90240154	924510	AB2-100 TAP	230.0	DVP	314583	6LAKEVE W	230.0	DVP	1	DVP_P1-2: LN 238-A	operation	375.06	129.82	131.74	AC	7.35

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-292	Upgrade Number
42958356	2	3BTLEBRO 115.0 kV - 3ROCKYMT115T 115.0 kV Ckt 1	DVP n6118: PJM Network Upgrade n6118: Upgrading the breaker leads at DVP's terminal will bring the rating to 239/239/239 MVA (Limited by terminal equipment at Rocky Mount) Dominion End Ratings: 438/478/581 MVA Project Type : FAC Cost : \$100,000 Time Estimate : 30-36 Months New Ratings: Rate A: 239 MVA Rate B: 239 MVA Rate C: 239 MVA Queue Project AF1-292 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-292 could receive cost allocation. Note 2: Although Queue Project AF1-292 may not have cost responsibility for this upgrade, Queue Project AF1-292 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-292 comes into service prior to completion of the upgrade, Queue Project AF1-292 will need an interim study. n6618: PJM Network Upgrade n6618: 1. Split the 155 kV Bus at Hathaway into two separate buses with a 115 kV Line on each bus. 2. Rebuild Line #55 (Tarboro – Anaconda) and close the tie switch between Line 55 & 80. 3. Line #1001 is opened at Battleboro thus making Line #1001 radial from Chestnut Substation Project Type : CON Cost : \$6,900,000 Time Estimate : 20 Months The upgrade will need to be tested during the Facilities Study phase to ensure it does not cause any additional violations. Queue Project AF1-292 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-292 could receive cost allocation. Note 2: Although Queue Project AF1-292 may not have cost responsibility for this upgrade, Queue Project AF1-292 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-292 comes into service prior to completion of the upgrade, Queue Project AF1-292 will need an interim study. DEP See Next Page	See Next Page	See Next Page	n6118, n6618

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-292	Upgrade Number
42958356	2	3BTLEBRO 115.0 kV - 3ROCKYMT115T 115.0 kV Ckt 1	<u>DEP (cont'd)</u> Note 1: Duke Energy/Progress Portion: A potential constraint was identified by PJM on the Duke Energy/Progress (DEP) portion of the Battleboro- Rocky Mountain 115 kV line. The Queue Project AF1-292 may be subject to operational restriction if real-time system reliability issues occur. Additionally, if a baseline Network Upgrade project is identified on the Battleboro- Rocky Mountain 115 kV line prior to the execution of the Queue Project's final agreements, the Queue Project may require this upgrade to be in-service to be deliverable to the PJM system. If Queue Project AF1-292 comes into service prior to completion of the baseline Network Upgrade, Queue Project AF1-292 will need an interim deliverability study Note 2: There are no mitigations currently planned for the DEP portion of this overload. Though, the following upgrade has been scoped by DEP to increase the DEP End Rating to 301/301/301. Note 3: The existing DEP rating on Battleboro – Rocky Mountain 115 kV line is 164/164/164 MVA. Scope: Reconductor the circuit to 1590 ACSR will bring the rating to 185/205/214 MVA. Upgrading the terminal equipment at Rocky Mount will bring the rating to 301/301/301 MVA (Limited by 1590 ACSR conductor). Project Type: FAC Cost: TBD Time Estimate: 36 Months	\$7,000,000 Excludes DEP Upgrade Costs	\$0	n6118, n6618
90239691	3	AB2-100 TAP 230.0 kV - 6LAKEVEW 230.0 kV Ckt 1	<u>DVP</u> b3121 : PJM Baseline Upgrade b3121. Rebuild Clubhouse-Lakeview 230 kV Line #254 with single-circuit wood pole equivalent structures at the current 230 kV standard with a minimum rating of 1047 MVA. The baseline project has a projected in-service date of 10/15/2023. Project Type : FAC Cost : \$27,000,000 Note 1: Although Queue Project AF1-292 may not have cost responsibility for this upgrade, Queue Project AF1-292 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-292 comes into service prior to completion of the upgrade, Queue Project AF1-292 will need an interim study.	\$0	\$0	b3121

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-292	Upgrade Number
90240042	1	6SAPONY 230.0 kV - 6CARSON 230.0 kV Ckt 1	DVP n6221: Rebuild 11.79 miles of 230 kV Line 238 from Sapony to Carson with 2-795 150C ACSR. Project Type : FAC Cost : \$17,685,000 Time Estimate : 30-36 Months New Ratings: Rate A: 1225 MVA Rate B: 1225 MVA Rate C: 1409 MVA Queue Project AF1-292 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-292 could receive cost allocation. Note 2: Although Queue Project AF1-292 may not have cost responsibility for this upgrade, Queue Project AF1-292 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-292 comes into service prior to completion of the upgrade, Queue Project AF1-292 will need an interim study.	\$17,685,000	\$0	n6221
TOTAL COST				\$0	\$0	

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
90240042	314435	6SAPONY	DVP	314282	6CARSON	DVP	1	DVP_P1-2: LN 2012-B	single	678.68	99.94	100.41	AC	4.03

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
314435	6SAPONY	1.8637	80/20	1.8637
314572	3EMPORIA	0.1914	80/20	0.1914
314623	3WITAKRS	0.2603	80/20	0.2603
314704	3LAWRENC	0.7065	80/20	0.7065
315126	1ROARAP2	2.5511	80/20	2.5511
315128	1ROARAP4	2.4345	80/20	2.4345
315131	1EDGECEMA (Deactivation : 22/04/2019)	9.8513	80/20	9.8513
315132	1EDGECEMB (Deactivation : 22/04/2019)	9.8513	80/20	9.8513
315136	1ROSEMG1	5.0085	80/20	5.0085
315137	1ROSEMS1	3.1058	80/20	3.1058
315138	1ROSEMG2	2.3473	80/20	2.3473
315139	1GASTONA	7.7296	80/20	7.7296
315141	1GASTONB	7.7296	80/20	7.7296
315158	1KERR 1	0.4202	80/20	0.4202
315159	1KERR 2	1.1767	80/20	1.1767
315160	1KERR 3	1.1767	80/20	1.1767
315161	1KERR 4	1.1767	80/20	1.1767
315162	1KERR 5	1.1767	80/20	1.1767
315163	1KERR 6	1.1767	80/20	1.1767
315164	1KERR 7	1.1767	80/20	1.1767
315606	3AA2-053SOLA	2.3631	80/20	2.3631
315607	3AA1-063SOLA	2.1157	80/20	2.1157
315612	3AA2-057SOLA	1.4589	80/20	1.4589
920591	AA2-165 C	0.1730	80/20	0.1730
920671	AA2-174 C OP	0.1082	80/20	0.1082
922922	AB1-081 C OP	9.3257	80/20	9.3257
923262	AB1-132 C OP (Suspended)	33.3270	80/20	33.3270
923572	AB1-173 C OP	5.1278	80/20	5.1278
923582	AB1-173AC OP	5.1278	80/20	5.1278
923911	AB2-031 C O1	5.0899	80/20	5.0899
923991	AB2-040 C O1 (Suspended)	16.7130	80/20	16.7130
924151	AB2-059 C OP	1.9466	80/20	1.9466
924301	AB2-077 C O1 (Suspended)	1.4601	Adder	1.72
924311	AB2-078 C O1 (Suspended)	1.4601	Adder	1.72
924321	AB2-079 C O1 (Suspended)	1.4601	Adder	1.72

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
924501	AB2-099 C (Suspended)	0.3478	Adder	0.41
924511	AB2-100 C	6.6923	80/20	6.6923
925171	AB2-174 C O1	2.9755	80/20	2.9755
925591	AC1-034 C	7.1192	80/20	7.1192
925611	AC1-036 C	0.5469	Adder	0.64
925781	AC1-054 C O1	8.4706	80/20	8.4706
926071	AC1-086 C	49.0780	80/20	49.0780
926201	AC1-098 C	6.7349	80/20	6.7349
926211	AC1-099 C	2.2569	80/20	2.2569
927021	AC1-189 C	5.4087	Adder	6.36
927141	AC1-208 C	11.4689	80/20	11.4689
932631	AC2-084 C	9.6008	80/20	9.6008
934331	AD1-057 C O1	16.5571	80/20	16.5571
936261	AD2-033 C	7.8652	Adder	9.25
936361	AD2-046 C O1	8.5554	80/20	8.5554
936401	AD2-051 C O1	20.5314	80/20	20.5314
936481	AD2-063 C O1	10.1454	Adder	11.94
939181	AE1-148 C O1	8.3986	80/20	8.3986
940241	AE2-006	0.2782	Adder	0.33
940481	AE2-033 C	65.5325	80/20	65.5325
940541	AE2-040 O1	11.9716	80/20	11.9716
940571	AE2-044 C	4.6628	80/20	4.6628
940661	AE2-053 O1	2.6440	Adder	3.11
941541	AE2-151 C	0.6021	Adder	0.71
942451	AE2-258	1.7378	Adder	2.04
942471	AE2-260 C O1	48.3679	80/20	48.3679
943171	AE2-346 C	0.8347	Adder	0.98
943911	AF1-059	28.2900	80/20	28.2900
944141	AF1-082	2.9975	80/20	2.9975
946011	AF1-266	54.8558	80/20	54.8558
946281	AF1-292 C	4.0332	80/20	4.0332
WEC	WEC	0.2775	Confirmed LTF	0.2775
LGEE	LGEE	0.5226	Confirmed LTF	0.5226
CPL	CPL	3.0700	Confirmed LTF	3.0700
CBM-W2	CBM-W2	10.8845	Confirmed LTF	10.8845
NY	NY	0.3290	Confirmed LTF	0.3290
TVA	TVA	2.2666	Confirmed LTF	2.2666
CBM-S2	CBM-S2	19.3919	Confirmed LTF	19.3919
CBM-S1	CBM-S1	12.6863	Confirmed LTF	12.6863
MADISON	MADISON	0.6088	Confirmed LTF	0.6088
MEC	MEC	1.6621	Confirmed LTF	1.6621
AA2-074	AA2-074	2.0925	LTF	2.0925
CBM-W1	CBM-W1	10.3958	Confirmed LTF	10.3958

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42958356	314554	3BTLEBRO	DVP	304223	3ROCKYMT115T	CPL	1	DVP_P7-1: LN 2058-2181	tower	93.0	453.18	454.13	AC	1.09

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
314541	3WATKINS	0.2104	Adder	0.25
314582	3KELFORD	0.3985	50/50	0.3985
314623	3WITAKRS	0.5684	50/50	0.5684
315131	1EDGECEMA (Deactivation : 22/04/2019)	12.1440	50/50	12.1440
315132	1EDGECEMB (Deactivation : 22/04/2019)	12.1440	50/50	12.1440
315136	1ROSEMG1	1.6358	50/50	1.6358
315137	1ROSEMS1	1.0143	50/50	1.0143
315138	1ROSEMG2	0.7666	50/50	0.7666
315139	1GASTONA	2.1022	50/50	2.1022
315141	1GASTONB	2.1022	50/50	2.1022
315601	1CONETOE2SOL	1.3631	50/50	1.3631
315612	3AA2-057SOLA	2.6249	50/50	2.6249
900672	V4-068 E	0.1238	Adder	0.15
917332	Z2-043 E	0.9783	50/50	0.9783
917342	Z2-044 E	1.3755	50/50	1.3755
917512	Z2-088 E OP1	3.2983	50/50	3.2983
918492	AA1-063AE OP	1.9435	Adder	2.29
918512	AA1-065 E OP	1.4302	Adder	1.68
918532	AA1-067 E	0.2465	Adder	0.29
919692	AA2-053 E OP	1.9785	Adder	2.33
919702	AA2-057 E OP	6.9959	50/50	6.9959
920042	AA2-088 E OP	4.0157	Adder	4.72
920591	AA2-165 C	0.3112	50/50	0.3112
920592	AA2-165 E	0.9615	50/50	0.9615
920672	AA2-174 E OP	0.2286	Adder	0.27
922922	AB1-081 C OP	22.9678	50/50	22.9678
922923	AB1-081 E OP	9.8434	50/50	9.8434
923262	AB1-132 C OP (Suspended)	9.0636	50/50	9.0636
923263	AB1-132 E OP (Suspended)	3.8844	50/50	3.8844
923572	AB1-173 C OP	1.0076	Adder	1.19
923573	AB1-173 E OP	0.4702	Adder	0.55
923582	AB1-173AC OP	1.0076	Adder	1.19
923583	AB1-173AE OP	0.4702	Adder	0.55
923801	AB2-015 C OP	2.5105	Adder	2.95
923802	AB2-015 E OP	2.0586	Adder	2.42
923911	AB2-031 C O1	1.0002	Adder	1.18
923912	AB2-031 E O1	0.4926	Adder	0.58

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
923991	AB2-040 C O1 (Suspended)	3.2841	Adder	3.86
923992	AB2-040 E O1 (Suspended)	2.6870	Adder	3.16
924151	AB2-059 C OP	4.7943	50/50	4.7943
924152	AB2-059 E OP	13.9448	50/50	13.9448
924501	AB2-099 C (Suspended)	0.2636	Adder	0.31
924502	AB2-099 E (Suspended)	0.1130	Adder	0.13
924512	AB2-100 E	2.0925	Adder	2.46
925122	AB2-169 E	1.7713	Adder	2.08
925172	AB2-174 E O1	2.7080	Adder	3.19
925591	AC1-034 C	17.5335	50/50	17.5335
925592	AC1-034 E	13.2270	50/50	13.2270
926071	AC1-086 C	13.3472	50/50	13.3472
926072	AC1-086 E	6.0748	50/50	6.0748
926201	AC1-098 C	8.4829	50/50	8.4829
926202	AC1-098 E	5.0537	50/50	5.0537
926211	AC1-099 C	2.8427	50/50	2.8427
926212	AC1-099 E	1.6695	50/50	1.6695
927021	AC1-189 C	6.0689	50/50	6.0689
927022	AC1-189 E	3.0231	50/50	3.0231
927141	AC1-208 C	13.5209	50/50	13.5209
927142	AC1-208 E	6.0039	50/50	6.0039
932631	AC2-084 C	12.0927	50/50	12.0927
932632	AC2-084 E	5.9561	50/50	5.9561
933991	AD1-023 C	3.9006	Adder	4.59
933992	AD1-023 E	2.1235	Adder	2.5
934331	AD1-057 C O1	10.1813	50/50	10.1813
934332	AD1-057 E O1	5.4311	50/50	5.4311
936401	AD2-051 C O1	3.1690	Adder	3.73
936402	AD2-051 E O1	1.3607	Adder	1.6
938171	AE1-026 C O1	9.8964	Adder	11.64
938172	AE1-026 E O1	2.6109	Adder	3.07
938221	AE1-035 C	0.7986	Adder	0.94
938222	AE1-035 E	0.3933	Adder	0.46
940571	AE2-044 C	11.4839	50/50	11.4839
940572	AE2-044 E	4.9217	50/50	4.9217
941541	AE2-151 C	0.5010	Adder	0.59
941542	AE2-151 E	0.2698	Adder	0.32
942471	AE2-260 C O1	4.9509	Adder	5.82
942472	AE2-260 E O1	7.0222	Adder	8.26
943171	AE2-346 C	0.6326	Adder	0.74
943172	AE2-346 E	0.2711	Adder	0.32
944141	AF1-082	7.3825	50/50	7.3825
946281	AF1-292 C	0.5550	Adder	0.65
946282	AF1-292 E	0.3742	Adder	0.44
NEWTON	NEWTON	0.8671	Confirmed LTF	0.8671
FARMERCITY	FARMERCITY	0.0497	Confirmed LTF	0.0497
G-007A	G-007A	0.4795	Confirmed LTF	0.4795
VFT	VFT	1.2707	Confirmed LTF	1.2707
PRAIRIE	PRAIRIE	2.3402	Confirmed LTF	2.3402
COFFEEN	COFFEEN	0.1620	Confirmed LTF	0.1620
CHEOAH	CHEOAH	0.7042	Confirmed LTF	0.7042

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
EDWARDS	EDWARDS	0.2618	Confirmed LTF	0.2618
TILTON	TILTON	0.4631	Confirmed LTF	0.4631
GIBSON	GIBSON	0.4193	Confirmed LTF	0.4193
CALDERWOOD	CALDERWOOD	0.6869	Confirmed LTF	0.6869
BLUEG	BLUEG	1.2933	Confirmed LTF	1.2933
TRIMBLE	TRIMBLE	0.4107	Confirmed LTF	0.4107
CATAWBA	CATAWBA	0.7406	Confirmed LTF	0.7406

11.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
90239691	924510	AB2-100 TAP	DVP	314583	6LAKEVEW	DVP	1	DVP_P4-2: 23872	breaker	459.0	106.1	107.67	AC	7.35

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
314572	3EMPORIA	0.1939	50/50	0.1939
314704	3LAWRENC	0.7347	50/50	0.7347
923572	AB1-173 C OP	4.0258	50/50	4.0258
923573	AB1-173 E OP	1.8787	50/50	1.8787
923582	AB1-173AC OP	4.0258	50/50	4.0258
923583	AB1-173AE OP	1.8787	50/50	1.8787
923911	AB2-031 C O1	3.9960	50/50	3.9960
923912	AB2-031 E O1	1.9682	50/50	1.9682
923991	AB2-040 C O1 (Suspended)	13.1212	50/50	13.1212
923992	AB2-040 E O1 (Suspended)	10.7356	50/50	10.7356
924022	AB2-043 E O1	1.3953	Adder	1.64
924162	AB2-060 E OP	2.0848	Adder	2.45
924301	AB2-077 C O1 (Suspended)	0.9558	Adder	1.12
924302	AB2-077 E O1 (Suspended)	0.6372	Adder	0.75
924311	AB2-078 C O1 (Suspended)	0.9558	Adder	1.12
924312	AB2-078 E O1 (Suspended)	0.6372	Adder	0.75
924321	AB2-079 C O1 (Suspended)	0.9558	Adder	1.12
924322	AB2-079 E O1 (Suspended)	0.6372	Adder	0.75
924511	AB2-100 C	9.1527	50/50	9.1527
924512	AB2-100 E	25.4529	50/50	25.4529
925171	AB2-174 C O1	2.6177	50/50	2.6177
925172	AB2-174 E O1	13.3722	50/50	13.3722
925611	AC1-036 C	0.3721	Adder	0.44
925612	AC1-036 E	0.6072	Adder	0.71
936261	AD2-033 C	5.3318	Adder	6.27
936262	AD2-033 E	3.5546	Adder	4.18
936481	AD2-063 C O1	6.8314	Adder	8.04
936482	AD2-063 E O1	4.5163	Adder	5.31
940241	AE2-006	0.1893	Adder	0.22
940481	AE2-033 C	67.1754	50/50	67.1754
940482	AE2-033 E	45.2868	50/50	45.2868
942451	AE2-258	1.1641	Adder	1.37
942471	AE2-260 C O1	62.4203	50/50	62.4203
942472	AE2-260 E O1	88.5357	50/50	88.5357
943911	AF1-059	27.4520	50/50	27.4520

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
946011	AF1-266	56.2311	50/50	56.2311
946281	AF1-292 C	4.3924	50/50	4.3924
946282	AF1-292 E	2.9612	50/50	2.9612
NEWTON	NEWTON	0.0655	Confirmed LTF	0.0655
FARMERCITY	FARMERCITY	0.0040	Confirmed LTF	0.0040
NY	NY	0.0155	Confirmed LTF	0.0155
PRAIRIE	PRAIRIE	0.1886	Confirmed LTF	0.1886
O-066	O-066	0.1882	Confirmed LTF	0.1882
COFFEEN	COFFEEN	0.0124	Confirmed LTF	0.0124
CHEOAH	CHEOAH	0.0636	Confirmed LTF	0.0636
EDWARDS	EDWARDS	0.0196	Confirmed LTF	0.0196
TILTON	TILTON	0.0334	Confirmed LTF	0.0334
G-007	G-007	0.0291	Confirmed LTF	0.0291
GIBSON	GIBSON	0.0306	Confirmed LTF	0.0306
CALDERWOOD	CALDERWOOD	0.0616	Confirmed LTF	0.0616
BLUEG	BLUEG	0.0903	Confirmed LTF	0.0903
TRIMBLE	TRIMBLE	0.0284	Confirmed LTF	0.0284
CATAWBA	CATAWBA	0.0742	Confirmed LTF	0.0742

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA1-063	Huntsville (Cabin Creek) 69kV	Withdrawn
AA1-063A	Carolina–Seaboard 115kV	In Service
AA1-065	Earleys 230kV	In Service
AA1-067	Everetts 34.5kV	In Service
AA2-053	Carolina-Jackson 115kV	In Service
AA2-057	Hornertown-Whitakers 115kV	In Service
AA2-074	CPLP-PJM	Confirmed
AA2-088	Boykins-Handsome 115kV	In Service
AA2-165	Hornertown-Whitakers 115kV	In Service
AA2-174	Carolina-Jackson 115kV	In Service
AB1-081	Anaconda-Mayo Dunbar 115kV	Under Construction
AB1-132	Thelma 230kV	Suspended
AB1-173	Brink-Trego 115kV	Engineering and Procurement
AB1-173A	Brink-Trego 115kV	Engineering and Procurement
AB2-015	Franklin 115kV	Engineering and Procurement
AB2-031	Brink-Trego 115kV	Engineering and Procurement
AB2-040	Brink 115kV	Suspended
AB2-043	Chase City 115kV	Under Construction
AB2-059	Benson-Dunbar 115kV	Under Construction
AB2-060	Chase City-Lunenburg 115kV	Under Construction
AB2-077	Buggs Island-Chase City 115kV	Suspended
AB2-078	Buggs Island-Chase City 115kV	Suspended
AB2-079	Buggs Island-Chase City 115kV	Suspended
AB2-099	Ahoskie 34.5kV	Suspended
AB2-100	Clubhouse-Lakeview 230kV	Under Construction
AB2-169	Pantago-Five Points 115kV	Partially in Service - Under Construction
AB2-174	Emporia-Trego 115kV	Under Construction
AC1-034	Heartsease DP - Mayo Dunbar 115kV	Active
AC1-036	Twittys Creek 34.5kV	Engineering and Procurement
AC1-054	Kerr Dam–Eatons Ferry 115 kV	Engineering and Procurement
AC1-086	Thelma 230kV	Active
AC1-098	Dawson-South Justice 115kV	Engineering and Procurement
AC1-099	Dawson-South Justice 115kV	Engineering and Procurement
AC1-189	Chinquapin-Everetts 230kV	Active
AC1-208	Cox-Whitakers 115kV	Active
AC2-084	Dawson-South Justice 115kV	Active
AD1-023	Cashie-Trowbridge 230 kV	Active
AD1-057	Hornertown-Hathaway 230 kV	Active
AD2-033	Chase City-Lunenburg 115 kV	Active

Queue Number	Project Name	Status
AD2-046	Boynton DP-Kerr Dam 115 kV	Active
AD2-051	Earleys – Northampton 230kV	Active
AD2-063	Central-Chase City 115kV	Active
AE1-026	Cashie 230 kV	Active
AE1-035	Earleys 230 kV	Engineering and Procurement
AE1-148	Kerr Dam-Ridge Rd 115 kV	Active
AE2-006	Twittys Creek 34.5 kV	Engineering and Procurement
AE2-033	Clubhouse-Sappony 230 kV	Active
AE2-040	Sapony 34.5 kV	Active
AE2-044	Anaconda-Dunbar 115 kV	Active
AE2-053	Kerr Dam-Ridge Road 115 kV	Active
AE2-151	Earleys 34.5kV	Engineering and Procurement
AE2-258	Chase City 115 kV	Active
AE2-260	Clubhouse 230 kV	Active
AE2-346	Ahoskie 34.5 kV	Active
AF1-059	Brodnax-South Hill 115 kV	Active
AF1-082	Heartsease-Mayo Dunbar DP	Active
AF1-266	Clubhouse-Sapony 230 kV	Active
AF1-292	Fields 34.5kV	Active
V4-068	Murphy's 34.5kV	In Service
Z2-043	Kelford 34.5kV	In Service
Z2-044	Whitakers 34.5kV	In Service
Z2-088	Tarboro-Everetts 230kV	In Service

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
DVP_P4-2: 23872	CONTINGENCY 'DVP_P4-2: 23872' /* CARSON 230 KV OPEN BRANCH FROM BUS 314282 TO BUS 314435 CKT 1 /* 6CARSON 230.00 - 6SAPONY 230.00 OPEN BRANCH FROM BUS 314435 TO BUS 940480 CKT 1 /* 6SAPONY 230.00 - AE2- 033 TAP 230.00 OPEN BUS 314435 /* ISLAND: 6SAPONY 230.00 OPEN BUS 923852 /* ISLAND: AB2-025 E 230.00 OPEN BRANCH FROM BUS 314282 TO BUS 314902 CKT 1 /* 6CARSON 230.00 - 8CARSON 500.00 OPEN BUS 314455 /* 6CARSO_1 230.00 KV END
DVP_P1-2: LN 2012-B	CONTINGENCY 'DVP_P1-2: LN 2012-B' OPEN BRANCH FROM BUS 936400 TO BUS 314569 CKT 1 /* AD2-051 TAP230.00 - 6EARLEYS 230.00 END
DVP_P1-2: LN 238-A	CONTINGENCY 'DVP_P1-2: LN 238-A' OPEN BRANCH FROM BUS 314282 TO BUS 314435 CKT 1 /* 6CARSON 230.00 - 6SAPONY 230.00 OPEN BRANCH FROM BUS 314435 TO BUS 940480 CKT 1 /* 6SAPONY 230.00 - AE2- 033 TAP 230.00 OPEN BUS 314435 /* ISLAND: 6SAPONY 230.00 OPEN BUS 923852 /* ISLAND: AB2-025 E 230.00 END
DVP_P7-1: LN 2058-2181	CONTINGENCY 'DVP_P7-1: LN 2058-2181' /* . OPEN BRANCH FROM BUS 304222 TO BUS 313845 CKT 1 /* 6ROCKYMT230T230.00 - 6HATHAWAY 230.00 OPEN BRANCH FROM BUS 313844 TO BUS 313845 CKT 2 /* 3HATHAWAY 115.00 - 6HATHAWAY 230.00 OPEN BUS 304226 /* ISLAND: 6PA-RMOUNT#4115.00 OPEN BRANCH FROM BUS 304226 TO BUS 314591 CKT 1 /* 6PA-RMOUNT#4230.00 - 6NASH 230.00 OPEN BRANCH FROM BUS 313845 TO BUS 314591 CKT 1 /* 6HATHAWAY 230.00 - 6NASH 230.00 OPEN BUS 314591 /* ISLAND: 6NASH 230.00 END
314435 6SAPONY 230 940480 AE2-033 TAP 230 1	CONTINGENCY '314435 6SAPONY 230 940480 AE2-033 TAP 230 1' OPEN BRANCH FROM BUS 314435 TO BUS 940480 CKT 1 END

12 Light Load Analysis

Not required for this project.

13 Short Circuit Analysis

No circuit breakers were identified as overdutied as part of this analysis.

14 Stability and Reactive Power

To be determined in the Facilities Study Phase.

15 Affected Systems

15.1 Duke Energy-Progress

Potential constraints were identified by PJM on the following Dominion – Duke Energy/Progress (DEP) tie lines. Enter into an Affected System Facilities Study agreement with Duke / Progress Energy (DEP) to determine what, if any, reinforcements are required on their system. The following facilities were identified in this report:

- Battleboro – Rocky Mt. 115 kV line

Attachment 1: One Line Diagram

